ARJUNA JEE 2027

Lecture -03 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A Different types of Questions on A.P.

B Common Terms

C

D



SEQUENCE & SERIES

Question [JEE Mains-2022]

Suppose $a_1, a_2, \ldots, a_n, \ldots$ be an arithmetic progression of natural numbers. If the ratio of the sum of first five terms to the sum of first nine terms of the progression is $5 : 17$ and $110 < a_{15} < 120$, then the sum of the first ten terms of the progression is equal to

Ans: B

A 290

B 380

C 460

D 510

Question [JEE Mains-2025]

The number of terms of an A.P. is even; the sum of all the odd terms is 24, the sum of all the even terms is 30 and the last term exceeds the first by $\frac{21}{2}$. Then the number of terms which are integers in the A.P. is:

Ans: A

A 4

B 10

C 6

D 8

Question

X^th – O.P.

The sum of n terms of two A.P.'s are in the ratio of $7n + 1 : 4n + 27$, find the ratio of their 11th terms.

AP_1 : a & d

AP_2 : A & D

$$\frac{S_n}{S_n} = \frac{7n+1}{4n+27}$$

$$\frac{\frac{n}{2}\{2a+(n-1)d\}}{\frac{n}{2}\{2A+(n-1)D\}} = \frac{7n+1}{4n+27}$$

$$\frac{a+\left(\frac{n-1}{2}\right)d}{A+\left(\frac{n-1}{2}\right)D} = \frac{7n+1}{4n+27}$$

$\frac{n-1}{2} = 10$

$n = 21$

put $n = 21$

$$\frac{T_{11}}{T_{11}} = \frac{a+10d}{A+10D}$$

$$\frac{a+10d}{A+10D} = \frac{7\times21+1}{4\times21+27} = \frac{148}{111}$$

Sigma Notation

$$\sum_{r=1}^{5} r = 1+2+3+4+5$$

$$\sum_{\alpha=0}^{3} (2\alpha+3) = 3+5+7+9$$

Question [JEE Mains-2023]

Let $S_1, S_2, S_3, \ldots\ldots, S_{10}$ respectively be the sum to 12 terms of 10 A.P.s whose first terms are $1, 2, 3, \ldots\ldots, 10$ and the common differences are $1, 3, 5, \ldots\ldots, 19$ respectively. Then $\sum_{i=1}^{10} S_i$ is equal to

$\sum_{i=1}^{10} S_i = S_1 + S_2 + S_3 \ldots\ldots + S_{10}$

(A) 7380

(B) 7220

(C) 7360

(D) 7260

12 terms

$d=1$ $AP_1 \rightarrow 1, 2, 3 \ldots\ldots$ Sum $\rightarrow S_1 = \frac{12}{2}\{2\times1 + 11\times1\}$

$d=3$ $AP_2 \rightarrow 2, 5, 8, 11 \ldots\ldots$ $S_2 = \frac{12}{2}\{2\times2 + 11\times3\}$

$d=5$ $AP_3 \rightarrow 3, 8, 13 \ldots\ldots$ $S_3 = \frac{12}{2}\{2\times3 + 11\times5\}$

$\vdots$

$d=19$ $AP_{10} \rightarrow 10$ $S_{10} = \frac{12}{2}\{2\times10 + 11\times19\}$

$$S_1 = 6\{2\times 1 + 11\times 1\}$$

$$S_2 = 6\{2\times 2 + 11\times 3\}$$

$$S_3 = 6\{2\times 3 + 11\times 5\}$$

$$\vdots$$

$$S_{10} = 6\{2\times 10 + 11\times 19\}$$

add

$$S_1+S_2+\ldots+S_{10} = 6\left\{2\underbrace{(1+2+3+\ldots+10)}_{\downarrow} + 11\underbrace{(1+3+5+\ldots+19)}_{\downarrow}\right\}$$

$$= 6\left(\cancel{2}\times\frac{10}{\cancel{2}}\{\underbrace{2\times 1+9\times 1}_{\hookrightarrow 11}\} + 11\times\frac{5}{\cancel{10}}\{\underbrace{2\times 1+9\times 2}_{\downarrow\ 20}\}\right)$$

$$= 6(110+1100) = 7260.$$

IMPORTANT NOTE

A.P.

Numbers when divided by 6 leaves remainder 5 = $6n + 5$, $n \in I$

Numbers when divided by 11 leaves remainder 4 = $11n + 4$, $n \in I$.

Numbers when divided by 9 leaves remainder 1 = $9n + 1$, $n \in I$.

Numbers when divided by 8 leaves remainder 7 = $8n + 7$, $n \in I$.

Question [JEE Mains-2019]

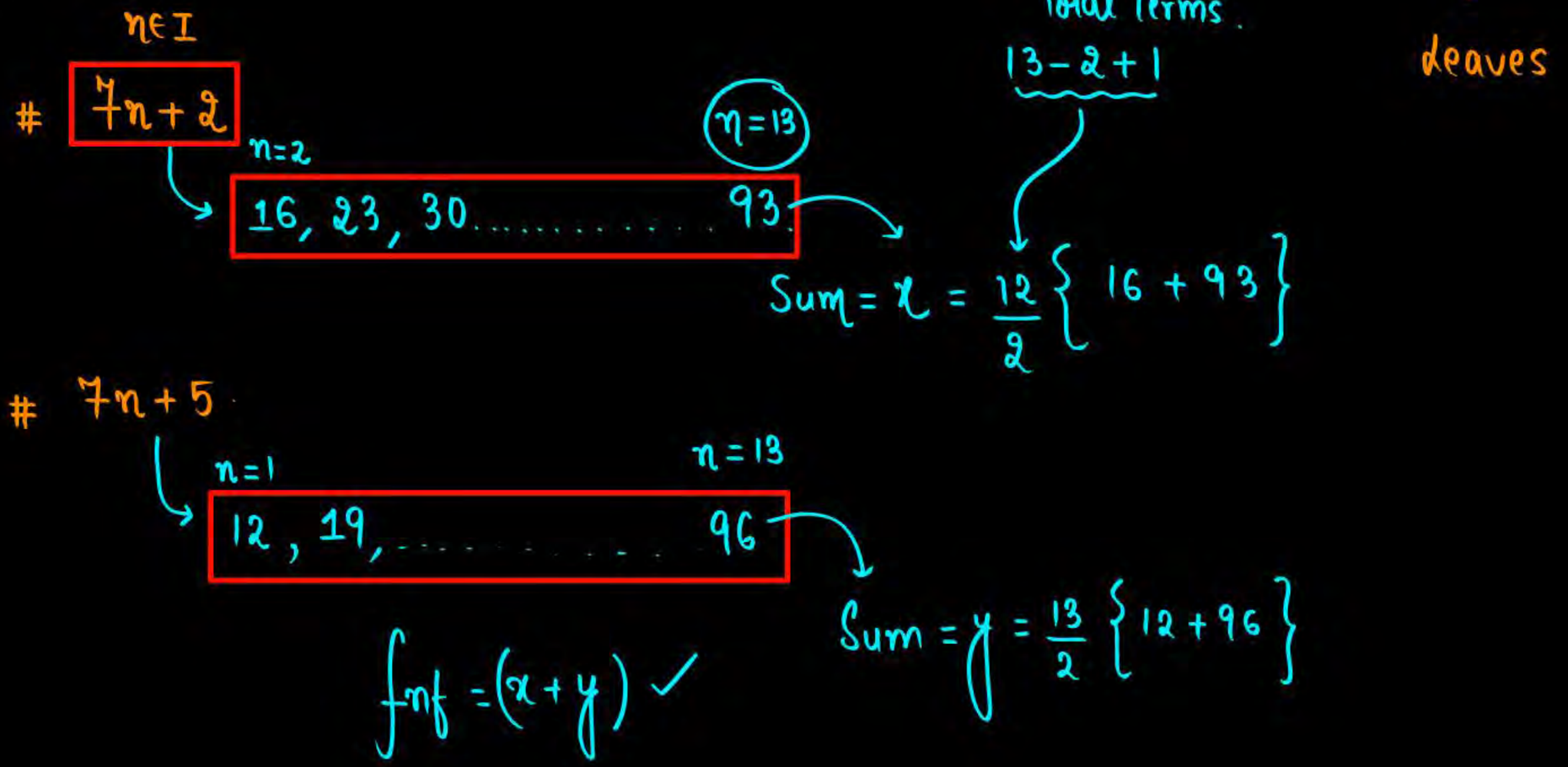
The sum of all two digit positive numbers which when divided by 7 yield 2 or 5 as remainder is

A 1256

B 1465

C 1365

D 1356



Question [JEE Mains-2023]

The sum of all those terms, of the arithmetic progression 3, 8, 13, ... 373, which are not divisible by 3, is equal to

[Ans. : (9525)]

3, 8, 13, 18, ... 373 → A.P.

$a = 3$

$d = 5, \ n = 75$.

3, 18, 33 ..... 363 → AP

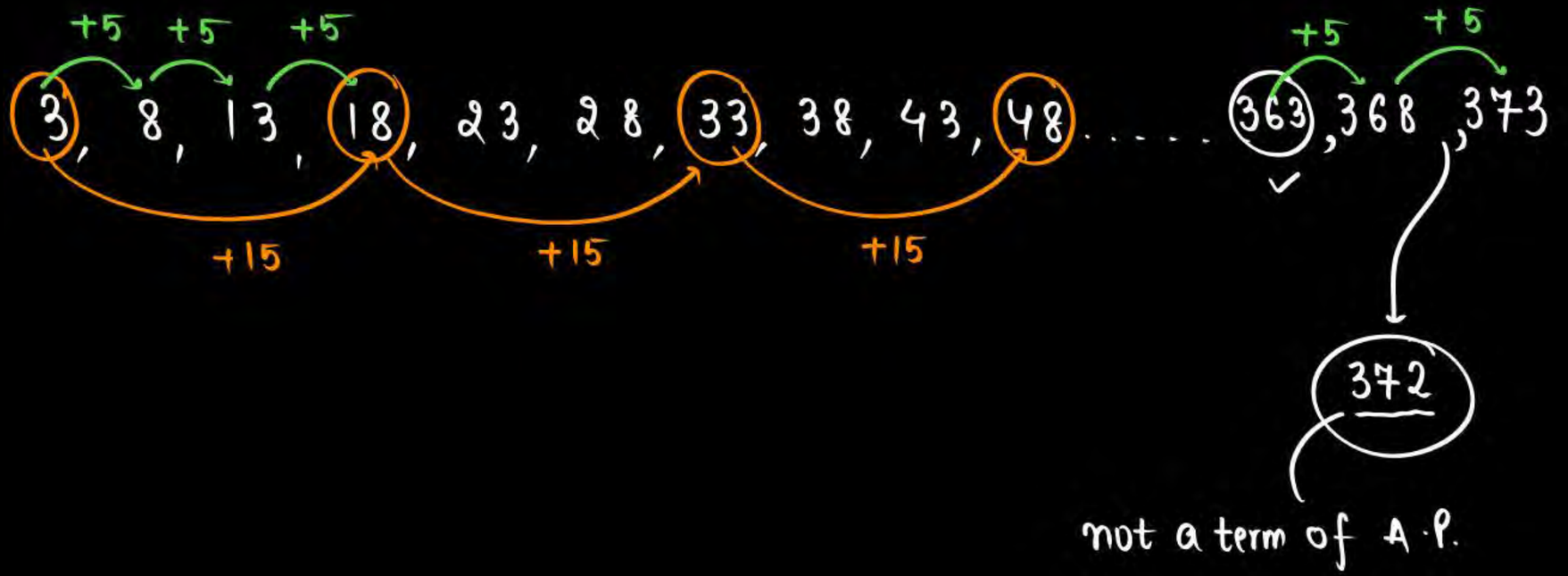
$a = 3$

$d = 15$

$n = 25$

Sum − (Sum of those terms which are div. by 3)

$$\text{Sum} = \left(\frac{75}{2}\{3 + 373\} - \frac{25}{2}\{3 + 363\} \right) = 9525.$$


+5 +5 +5
3, 8, 13, 18, 23, 28, 33, 38, 43, 48 . . . . . 363, 368, 373
+5 +5
+15 +15 +15
372
not a term of A.P.

Question

Consider set $\{1, 2, 3, 4, \ldots 200\}$ then number of numbers in given set which are:

(a) multiples of 2 are $A = \{2, 4, 6, 8, \ldots 200\} \Rightarrow (100)$

(b) multiples of 3 are $B = \{3, 6, 9, 12, \ldots 198\} \Rightarrow (66)$

(c) multiples of 2 or 3 are (133) .

(d) neither multiple of 2 nor multiple of 3 are 67 .

AND

multiple of 2 & 3

OR

multiple of 6 = $\{6, 12, 18, \ldots 198\}$ → (33)

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$= 100 + 66 - 33$$

$$= 133.$$

Total elements − $(n(A \cup B))$

↓

$200 - 133 = 67.$

Question [JEE Mains-2024]

For $A = \{n \in [100, 700] \cap \mathbb{N}, n$ is neither a multiple of 3 nor a multiple of 4$\}$. Then the number of elements in A is

A 290

B 280

C 300

D 310

$\{100, 101, 102, \ldots\ldots 700\}$

neither multiple of 3 nor 4.

total

Question [JEE Mains-2022]

Sawal puchne ka tarika thoda casual hai.

The sum of all the elements of the set

$\{\alpha \in \{1, 2, \ldots\ldots, 100\} : \text{HCF}(\alpha, 24) = 1\}$ **is _____.**

G.C.D.

$A = \{2, 4, 6, 8 \ldots 100\} \Rightarrow n(A) = 50.$

$B = \{3, 6, 9 \ldots 99\} \Rightarrow n(B) = 33.$

$A \cap B = \{6, 12, 18 \ldots\ldots, 96\} \Rightarrow n(A \cap B) = 16.$

neither multiple of 2 nor 3 = Total − $(n(A \cup B))$

$= 100 - (50 + 33 - 16)$

DIBY - (X)

$\text{fnf} = (1+2+3+\ldots+100) - \{(2+4+6+\ldots+100) + (3+6+\ldots+99) - (6+12+\ldots 96)\}$

$\text{HCF}(\alpha, 24) = 1$

\# neither multiple of 2 nor 3.

$\text{HCF}(??, 2^3 3^1) = 1$

$\text{HCF}(2^1 5^1, 2^3 3^1) = 2^1 3^0 5^0 = 2$

$2^1 3^0 5^1 \quad 2^3 3^1 5^0$

$\text{HCF}(5^1 7^1, 2^3 3^1) = 2^0 3^0 5^0 7^0 = 1$ ✓

$2^0 3^0 5^1 7^1 \quad 2^3 3^1 5^0 7^0$

$\text{HCF}(2^2 3^2, 2^3 3^1) = 2^2 3^1 = 12$

$\text{HCF}(2^0 5^3 7^3, 2^3 3^1 5^1 7^0) = 2^0 3^0 5^0 7^0 = 1$ ✓

α and 24 are coprime

OR

$\text{GCD}(\alpha, 24) = 1$

Note :

LCM

$LCM(3,4) = 12$

सबसे छोटा नं० जो 3 & 4 दोनों की table में आता है।

$LCM\left(2^{3}3^{1}5^{2}, 2^{1}3^{2}5^{1}\right) = 2^{3}3^{2}5^{2}$

HCF / GCD

$HCF(3,4) = 1$

सबसे बड़ा नं० जिसकी Table में 3 & 4 आते हों।

$HCF\left(2^{3}3^{1}5^{2}, 2^{1}3^{2}5^{1}\right) = 2^{1}3^{1}5^{1}$

$LCM\left(2^{0}3^{1}, 2^{2}3^{0}\right) = 2^{2}3^{1} = 12$

Question [JEE Mains-2021]

[Ans. : (1000)]

The total number of 4-digit numbers whose greatest common divisor with 18 is 3, is

G.C.D.

{1000, 1001, 1002, ... 9999}

Ans.

Question

If a, b, c are real numbers satisfying the equation $25(9a^2 + b^2) + 9c^2 - 15(5ab + bc + 3ca) = 0$, then prove that a, b, c are in A.P.



$$-15\times5ab - 15bc - 15\times3ca$$

$$(5\times3\times a)^2 + (5b)^2 + (3c)^2 - (15a)(5b) - (5b)(3c) - (15a)(3c) = 0$$

$$x^2 + y^2 + z^2 - xy - yz - xz = 0$$

$$(x-y)^2 + (y-z)^2 + (z-x)^2 = 0$$

$$x = y = z$$

Question

If a, b, c are real numbers satisfying the equation $25(9a^2 + b^2) + 9c^2 - 15(5ab + bc + 3ca) = 0$, then prove that a, b, c are in A.P.

$x = y = z$

$15a = 5b = 3c$

$a = \frac{5}{15}b \qquad 5b = 3c$

$a = \frac{b}{3} \qquad \frac{5b}{3} = c$

HMP.

$2b = a + c$

$\frac{b}{3} + \frac{5b}{3}$

$\frac{6b}{3} = 2b = RHS.$

COMMON TERMS

100% Ques^{n} आएगा

Question [JEE Mains-2020] (1)

The number of terms common to the two A.P.'s 3, 7, 11, ..., 407 and 2, 9, 16, ...709 is_____.

$d_1 = 4$

AP_1 : 3, 7, 11, 15, 19, 23 ..., 407

AP_2 : 2, 9, 16, 23, 30, 37 ....., 709

$d_2 = 7$

C.T's (A.P) : 23, 51, 79, ... T_m

+28 +28

$d = LCM(d_1, d_2)$
$= LCM(4, 7) = 28$.

by observ.

T_{14}

14 CT's.

$T_n = 23 + (n-1)28$

$= 28n - 5$

$T_n \leq 407$

$T_n \leq 407$

$28n - 5 \leq 407$

$\# \quad n \leq \frac{412}{28}$

103

7

$\# \quad n \leq 14.\overline{m}$

$\boxed{n = 14}$

Question [JEE (Adv.)-2019] (2)

Let AP $(a; d)$ denote the set of all terms of an infinite arithmetic progression with first term a and common difference $d > 0$.
If AP : (1; 3) $\cap$ AP (2; 5) $\cap$ AP(3; 7) = AP $(a; d)$, then $a + d$ equals ___.

$\cap$ {
AP_1 : 1, 4, (7), 10, 13, 16, 19 ..... ∞
AP_2 : 2, (7), 12, 17, 22, 27 ..... ∞
AP_3 : 3, 10, 17, 24, 31, 38 ..... ∞
} $\Rightarrow \cap$ CT's (AP) : 7, 22, 37, ...

+15 +15

$d = LCM(3,5) = 15$

Question [JEE (Adv.)-2019] (2)

Let AP $(a; d)$ denote the set of all terms of an infinite arithmetic progression with first term a and common difference $d > 0$.
If AP : (1; 3) $\cap$ AP (2; 5) $\cap$ AP(3; 7) = AP $(a; d)$, then $a + d$ equals 157.

fnf

$\cap$ { AP : 7, 22, 37, 52, 67 ..... ∞ $(d = 15)$
AP$_3$: 3, 10, 17, 24, 31, 38, 45, 52 ... ∞ $(d = 7)$

CT's (A.P) : 52, , , ... = AP $(\overset{a}{52}, \overset{d}{105})$
+105 +105

$d = LCM(15, 7)$
$= 105$

Question [JEE Mains-2023]

The sum of the common terms of the following three arithmetic progressions.

$3, 7, 11, 15, \ldots\ldots\ldots\ldots, 399$

$2, 5, 8, 11, \ldots\ldots\ldots\ldots\ldots\ldots\ldots 59$ and

$2, 7, 12, 17, \ldots\ldots\ldots, 197$, is equal to



80% QIBY

Question [JEE (Adv.)-2018, Paper-1]

Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, ..., and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, ... Then, the number of elements in the set $X \cup Y$ is _____.

HW.
(809. Q1BY)

GEOMETRIC PROGRESSION

first term = a

T_1 T_2 T_3 T_4

$a, ar, ar^2, ar^3, \ldots\ldots\ldots\ldots$ → G.P.

$\times r$ $\times r$ $\times r$

r = Common Ratio $= \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3} = \frac{T_5}{T_4} = \ldots\ldots$

$\#\ (r \neq 0)$

n^{th} Term $= T_n = a \cdot r^{n-1}$

• RAPID FIRE

$a_n = a r^{n-1}$

(i) $\langle a_n \rangle$ is G.P.

$a = 3$, $r = 7$

Then $a_{10} = ar^9 = 3(7)^9$. ✓

(ii) If $a_7 = 28$ & $a_{28} = 7$

then $r =$ _____

where $\langle a_n \rangle$ is G.P. → a, r

$a_7 = ar^6 = 28$

$a_{28} = ar^{27} = 7$

$\Rightarrow \frac{ar^6}{ar^{27}} = \frac{28}{7} = 4$

$\frac{1}{r^{21}} = 4$

$\frac{1}{4} = r^{21}$

$\left(\frac{1}{4}\right)^{\frac{1}{21}} = r$

(iii) $\langle a_n \rangle$ follows → G.P.

$a_n \cdot a_{n-2} = a_{n-1}^2$, $n \geq 3$ and

$a_3 = 8$, $a_5 = 16$, then a_{11} _____.

Sol^n:

$a_n \cdot a_{n-2} = a_{n-1} \cdot a_{n-1}$

$\frac{a_n}{a_{n-1}} = \frac{a_{n-1}}{a_{n-2}}$, $n \geq 3$.

$n=3$

$\frac{a_3}{a_2} = \frac{a_2}{a_1} = \frac{a_4}{a_3} = \frac{a_5}{a_4} = \cdots$ ✓

$x^{21} = \frac{1}{4} = 2^{-2}$

$$x^{21} = 2^{-2}$$

$$x = \left(2^{-2}\right)^{\frac{1}{21}} = \left(\frac{1}{4}\right)^{\frac{1}{21}}$$

$a^m = b$

$a = b^{\frac{1}{m}}$ $m \in$ odd

$a = \pm b^{\frac{1}{m}}$, $m \in$ even.

(iii) $\langle a_n \rangle$ follows G.P.

$a_n \cdot a_{n-2} = a_{n-1}^2$, $n \geq 3$ and

$a_3 = 8$, $a_5 = 16$, then a_{11} ——— .

Sol^n:

$$a_n \cdot a_{n-2} = a_{n-1} \cdot a_{n-1}$$

$$\frac{a_n}{a_{n-1}} = \frac{a_{n-1}}{a_{n-2}}, \quad n \geq 3.$$

$n=3$

$$\frac{a_3}{a_2} = \frac{a_2}{a_1} = \frac{a_4}{a_3} = \frac{a_5}{a_4} = \text{----} \checkmark$$

G.P.

↳ a, r.

$$a_3 = ar^2 = 8$$

$$a_5 = ar^4 = 16$$

$$\# \quad \frac{1}{r^2} = \frac{1}{2}$$

$$\# \quad r^2 = 2 \rightarrow r = \pm\sqrt{2}$$

$a(2) = 8$

$\# \ a = 4$

$$a_{11} = ar^{10} = 4 \times (2)^5 = 2^7 = 128.$$

↳ $(r^2)^5$

QUESTION PRACTICE

Challenger Question [JEE Mains-2023]

Let $a_1, a_2, a_3, \ldots\ldots$ be in an arithmetic progression of positive terms.

Let $A_k = a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots. + a_{2k-1}^2 - a_{2k}^2$.

If $A_3 = -153, A_5 = -435$ and $a_1^2 + a_2^2 + a_3^2 = 66$, then $a_{17} - A_7$ is equal to



Question [JEE Mains-2023]

For three positive integers p, q, r, $x^{pq^2} = y^{qr} = z^{p^2r}$ and $r = pq + 1$ such that $3, 3\log_y x, 3\log_z y, 7\log_x z$ are in A.P. with common difference. Then $r - p - q$ is equal to

A 2

B 6

C 12

D −6



DO IT BY YOURSELF (DIBY)

DIBY-09

If $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to

[JEE Mains-2021]

[Ans. : 3]

DIBY-10

If the 10th term of an A.P. is $\frac{1}{20}$ and its 20th term is $\frac{1}{10}$, then the sum of its first 200 terms is

[JEE Mains-2020]

(A) 100 (B) $50\frac{1}{4}$ (C) $100\frac{1}{2}$ (D) 50

[Ans. : (C)]

DIBY-11

If the sum of the series $20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} + \cdots$ upto n^{th} term is 488 and the nth term is negative, then

[JEE Mains-2020]

(A) $n = 41$

(B) n^{th} term is $-4\frac{2}{5}$

(C) $n = 60$

(D) n^{th} term is -4

[Ans. : (D)]

DIBY-12

Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $\frac{-1}{2}$, then the greatest number amongst them is

[JEE Mains-2020]

(a) $\frac{21}{2}$

(B) 16

(C) 27

(D) 7

[Ans. : (B)]

DIBY-13

If $3^{2\sin 2\alpha-1}, 14$ and $3^{4-2\sin 2\alpha}$ are the first three terms of an A.P. for some α, then the sixth term of this A.P. is: [JEE Mains-2020]

(A) 65 (B) 78 (C) 81 (D) 66

[Ans. : (D)]

DIBY-14

If $a_1, a_2, a_3.$ and $b_1, b_2, b_3.$ are in A.P., and $a_1 = 2, a_{10} = 3,\ a_1 b_1 = 1 = a_{10} b_{10}$, then $a_4 b_4$ is equal to. [JEE Mains-2022]

(A) $\frac{35}{27}$ (B) 1 (C) $\frac{27}{28}$ (D) $\frac{28}{27}$

[Ans. : (D)]

DIBY-15

A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of m is equal to: [JEE Mains-2024]

(A) 125 (B) 150 (C) 180 (D) 160

[Ans. : (B)]

DIBY-16

Consider two sets A and B, each containing three numbers in $A.P$. Let the sum and the product of the elements of A be 36 and p respectively and the sum and the product of the elements of B be 36 and q respectively. Let d and D be the common differences of AP's in A and B respectively such that $D = d + 3, d > 0$. If $\frac{p+q}{p-q} = \frac{19}{5}$, then $p - q$ is equal to [JEE Mains-2025]

(A) 600 (B) 450 (C) 630 (D) 540

[Ans. : (D)]

DIBY-17

Consider an A. P. of positive integers, whose sum of the first three terms is 54 and the sum of the first twenty terms lies between 1600 and 1800. Then its 11th term is: [JEE Mains-2025]

(A) 84 (B) 90 (C) 108 (D) 122

[Ans. : (B)]

DIBY-18

Number of 4 digit integers less than 2800 which are either divisible by 3 or by 11 is equal to ?

[JEE Mains-2023]

[Ans. : 710]

$\sum_{i=1}^{\left(\frac{n}{2}\right)} a_{2i} = a_2 + a_4 + a_6 + \ldots a_n = 120$

DIBY-19

If where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{i=1}^{n} a_i = 192, \sum_{i=1}^{n/2} a_{2i} = 120$,, then n is equal to [JEE Mains-2022]

(A) 48 (B) 96 (C) 92 (D) 104

[Ans. : (B)]

$a_1 + a_2 + a_3 + \ldots + a_n = 192$

DIBY-20

Let S_1 be the sum of first $2n$ term of an arithmetic progression. Let S_2 be the sum of first $4n$ terms of the same arithmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first $6n$ terms of the arithmetic progression is equal to

[JEE Mains-2021]

(A) 5000 (B) 1000 (C) 7000 (D) 3000

[Ans. : (D)]

DIBY-21

The sum of all natural numbers 'n' such that $100 < n < 200$ and H.C.F. $(91, n) > 1$ is

[JEE Mains-2019]

(A) 3221 (B) 3121 (C) 3203 (D) 3303

[Ans. : (B)]

DIBY-22

The series of positive multiples of 3 is divided into sets : {3}, {6, 9, 12}, {15, 18, 21, 24, 27}, ... Then the sum of the elements in the 11th set is equal to

[JEE Mains-2022]

[Ans. : 6993]

DIBY-23

Let $a_1, a_2, \ldots a_{30}$ be an A.P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{2i-1}$. If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to

[JEE Mains-2019]

(A) 52 (B) 57 (C) 47 (D) 42

[Ans. : (A)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module:

Exercise-02(Prabal): COMPLETE

Exercise-03(Setu): COMPLETE

It is never about end result

It's always about Journey

Thank You

#futureIITians